Mileage forecast + erroneous-report flags

One prediction table for all 656,601 test rows: forecast, cold start and error flags

1,396 km

Forecast MAE, 133 days

backtest on the real test fleet
vs 1,549 km best rule, 5,053 km naive

590-666 km

Nowcast MAE, up to 28 days

10-13% better than the best rule

99.3%

Recall on injected errors

5,338 of 493,554 reports flagged (1.1%)

163,047

forecast rows filled

159,302 model + 3,745 cold start
(149 new cars without history)

0 / 1

error flag per reported mileage

rule violations + deviation from
expected usage, with score and reason

1 file

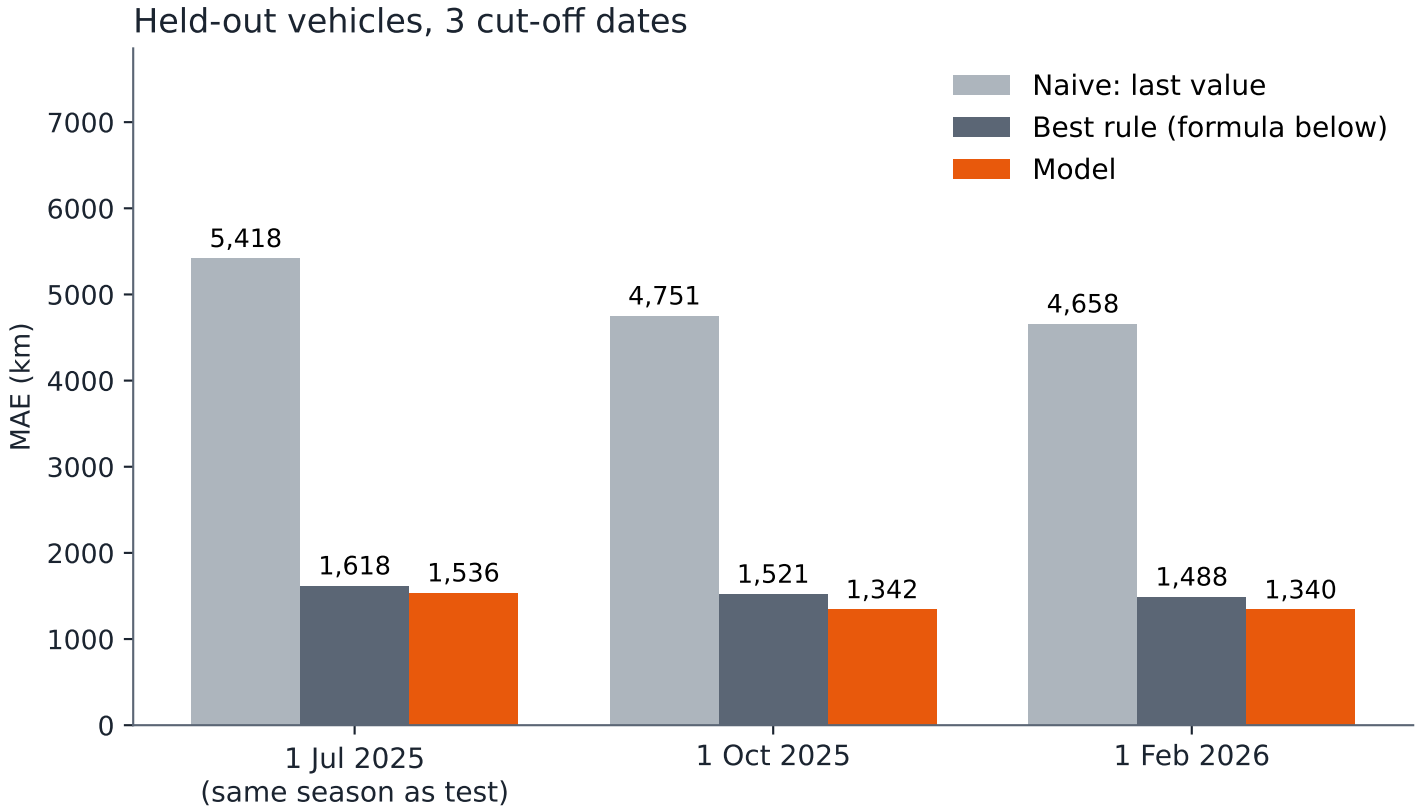
submission/predictions.parquet

same row order as test_data.parquet,
key: row_id

All accuracy numbers are measured on vehicles held out from training and tuning (no labels exist for the hidden test period).

Forecast: 133 days ahead

Gradient boosting on the mileage increment since the last known reading; error in km (MAE), lower is better



How the comparison rules predict

Naive: prediction = last known mileage

Best rule: prediction = last known mileage + class rate x days since the last reading

class rate = median km/day of the vehicles of the same class (Standard, Intermediate, Full Size) over the 90 days before the cut-off date

Model beats the best rule by

5.0%	2025-07-01
11.8%	2025-10-01
10.0%	2026-02-01

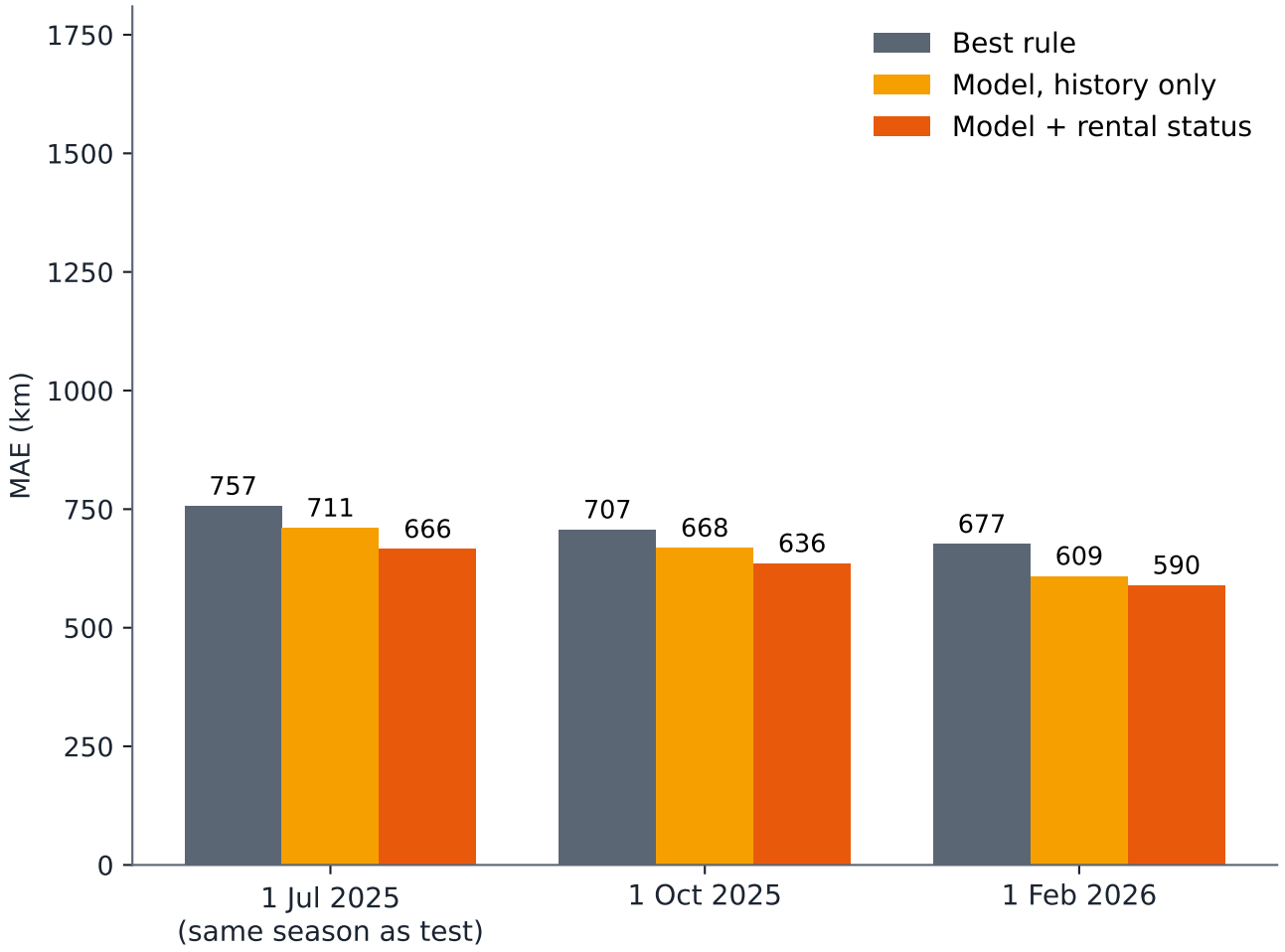
Backtest on the real test fleet

forecast from 18 Feb 2026, compared with the readings known until 1 Jul 2026 (same 133-day setup)

Naive (last value)	**5,053**
Best rule	**1,549**
Model (final)	**1,396**
Model + test history (not used)	**1,399**

Nowcast and new vehicles

Nowcast: mileage on a day without report (up to 28 days), using the rental status of that day



Cold start: 149 new cars with no mileage history

start at a typical new-car reading (about 12 km) and follow the median growth curve of new vehicles by age in days

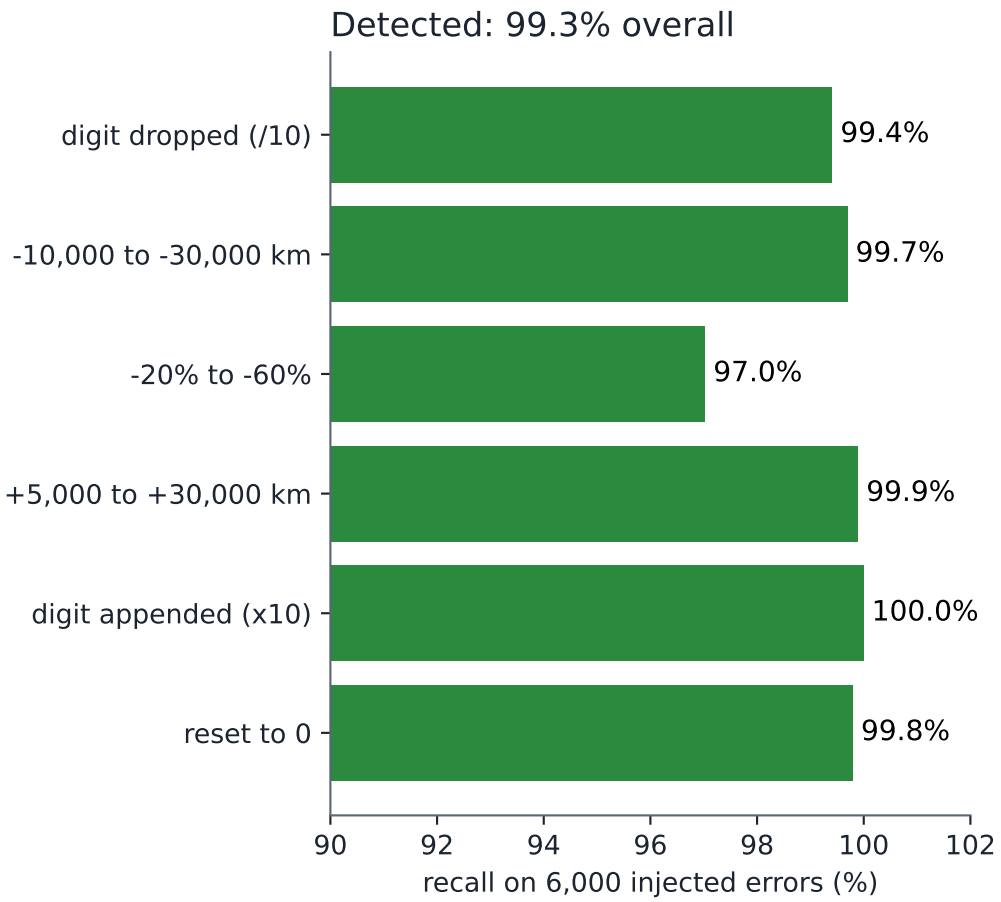
MAE on simulated new cars (km)

cut-off	fleet rate	age curve
2025-07-01	1,312	**1,255**
2025-10-01	1,091	**1,091**
2026-02-01	1,137	**961**

The age curve is never worse than the fleet-rate rule and cuts the error by 15% at one cut-off.

Erroneous-report flags

A report is flagged when it breaks physics or deviates strongly from expected usage



Five flags, combined with OR

rollback: reading below the previous one (longest non-decreasing subsequence marks the outlier itself)

too high: above all later readings

speed: more than 1,500 km/day between readings

high / low: increase far above or below the expected usage for that time gap (quantile bands per gap)

On the test file

5,338 of 493,554 reports flagged (1.1%)

2,781 by rules only, 2,257 by deviation only, 300 by both

every flag has a score and a plain-text reason

Injected errors are large; subtle manipulation is not covered by this test. No ground-truth labels exist.

Deliverables, limits, next steps

What is in the repo

submission/predictions.parquet: forecast + error flag for every test row

models/: final forecast model and nowcast model (joblib)

results/: all evaluation tables; reports/figures/: feature importance

src/ + scripts/: full pipeline, run with python -m scripts.run_<step>

SUBMISSION.md: approach, results, how to run

Next steps

Seasonal index and fleet trend as forecast features

Out-of-fold nowcast as expected value for the flags,
with quantile uncertainty bands

Cross-validation over vehicles

Checks with subtle manipulation (a few percent)

Known limits

Model underestimates in the summer cut-off (bias -516 km): no seasonal feature yet

Gain over the best rule varies by holdout and date: 5% to 12%

Error flags use a simple expected rate, not the trained model; no labels to verify precision

Nowcast is not filled per test row (empty test rows have no rental status)